# Design of a Temperature-Compensated PLL Frequency Synthesiser

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**Guide's Signature with Date**

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# Introduction & Problem Recap

- **Problem recap**
  - A PLL's ring-VCO drifts with temperature -- threshold voltage, mobility and capacitance all shift, so the same control voltage gives a different frequency hot vs. cold.
  - A compensation network cancels this by design, but sizing it needs a drift figure measured on the real topology, in the real process -- not assumed.
- **Objectives**
  - Design an integer-N charge-pump PLL (ref -> PFD -> charge pump -> loop filter -> 5-stage current-starved ring VCO -> /8 divider) in sky130, open-source, headless.
  - Measure uncompensated drift across all 5 process corners, then design and validate a temperature-compensation network with real statistical rigor (Monte Carlo + hypothesis testing).
- **Refinements after Review-I feedback**
  - Full 5-corner grid run (tt/ff/ss/sf/fs), not a 3-corner spot-check -- corrected the reported tempco spread from 3.4x to the real 12.5x (112-1399 ppm/C).
  - Cold-start acquisition risk (flagged as open in Review 1) fixed and verified: a true 0V cold start now locks in 839 ns.
  - Added a second-order (curvature-matched) compensation network, novelty-checked against prior art before committing to the claim (see Literature Review).

# Detailed Literature Review

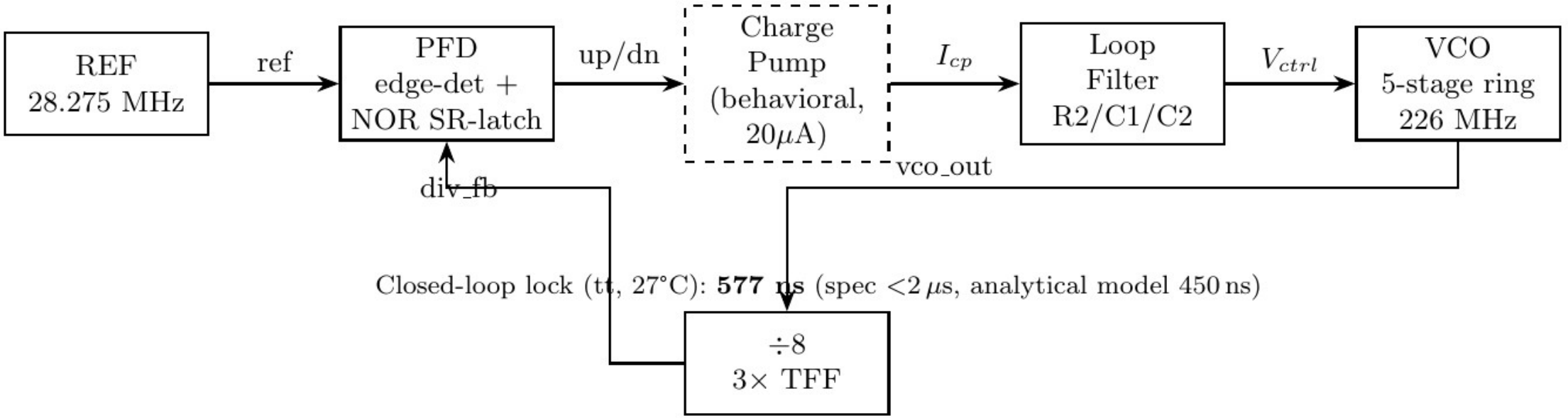
| No. | Author(s) & Year | Title / Source | Method / Approach | Findings & Research Gap |
|---|---|---|---|---|
| 1 | G. Souliotis et al., 2022 | Temperature compensated ring oscillator based VCO, AEU Int. J. Electron. Commun. | PTAT current compensation of a ring VCO inside a PLL; 180nm, 480MHz, fabricated | Closest topology benchmark: -1.5%/+0.4% over -40->120C, measured. First-order only; different node |
| 2 | C. Stancu et al., 2023 | Temp. and power-supply compensated CMOS clock circuit based on ring osc., Electronics | Temp + supply compensated 50MHz ring-oscillator clock; simulation only | 1.6% temperature error pre-layout. A clock circuit, never closed into a PLL |
| 3 | Z. Geng et al., 2018 | Low power fast locking PLL freq. synthesizer with temp. compensation, EAME | Freq. presetting + dynamic loop BW + static/dynamic temp. comp.; 180nm, 2.1-2.6GHz | Lock <2us over 0-80C, -117dBc/Hz@1MHz, 9mW. Stability claim is the locked output, set by the crystal |
| 4 | Y. You et al., 2014 | 12GHz 67% tuning, 0.37ps RJrms PLL, LC-VCO temp. comp., 0.13um CMOS, IEEE RFIC | LC-VCO PLL for serial links; comp. scheme preserves tuning range; 5.6-13.4GHz | Stays locked -40->85C. Needs inductor models our process supports poorly; not portable to a ring VCO |
| 5 | S. Sinha, 2002 | Design of an integrated CMOS PLL frequency synthesizer, IEEE MELECON | Dual-loop CP-PLL, 0.35um, 2.4-2.5GHz, 1MHz steps | -82dBc/Hz@10kHz. Architectural reference for our loop; no temperature compensation |
| 6 | E. M. V. Freitas et al., 2025 | Using open source EDA tools in ASICs for HEP: a mixed comparison, JINST | Open-source vs commercial flows on IHP 130nm blocks incl. a current-starved ring VCO | Digital QoR quantified (+53% area, +118% power); analog VCO comparison qualitative, open-loop, no MC |
| 7 | M. Thiriloganathan et al., 2026 | 2.4GHz LC-VCO fractional-N PLL, open-source, 130nm BiCMOS, SMACD | Fully open-source flow (Xschem/ngspice/KLayout/OpenEMS), EM-modelled inductor | -100.8dBc/Hz@1MHz. Closes a loop open-source, but no temp. compensation, no PVT/Monte Carlo |
| 8 | R. N. S. Raphael et al., 2019 | PLL-based frequency synthesizer for digital systems driving, BTSym'18 (Springer) | 3rd-order CP-PLL for clock generation, integer-N (N=8-16), VCO input-cap-aware LF model | Settling <2us. Same loop class as ours; no temperature compensation |

# Detailed Literature Review (contd.)

| No. | Author(s) & Year | Title / Source | Method / Approach | Findings & Research Gap |
|---|---|---|---|---|
| 9 | Y. Li, K. Du, J. Zhang, 2022 | Design of ring oscillator with temperature compensation effect, Int. J. RF Microw. CAE | Positive-TC supply voltage on a 31-stage ring, 0.18um; 33kHz sleep timer | 120 ppm/C (1.6%) over -40->80C. Single linear coefficient; kHz timer, not a PLL |
| 10 | H. H. Hammam et al., 2022 | An ultra-low-power process-and-temperature compensated ring oscillator, IEEE ICEEE | 3-stage current-starved ring, 130nm, process + temperature compensation | Up to 90% less variation vs. plain current-starved ring, 346nW. Standalone oscillator |
| 11 | S. Zhang et al., 2016 | Temperature compensation technique for ring oscillators with tail current, Electron. Lett. | Tail current = constant source + PTAT current, first-order | Foundational first-order technique; no numbers reported in abstract; not closed-loop |
| 12 | M. Ortner, M. Punzenberger, 2017 | Oscillators using quadratic temperature compensation, US Pat. 9,748,962 B2 | Linear + quadratic temperature term for a VCO, -40->85C | Generic 2nd-order compensation is patented -- narrows our claim to ring-VCO + closed PLL + open PDK + stats |
| 13 | D. Patrinos, G. Souliotis, 2025 | Low-voltage, low-power 2.5GHz ring oscillator with process & temp. comp., JLPEA | PTAT current + process-corner detection/tuning; 22nm, 0.8V; post-layout sim | Drift 23.9%->5.4% (tt) and 47.3%->7.2% (PVT). Corner-aware like our C(corner); not in a PLL |
| 14 | M. Sivasakthi, P. Radhika, 2024 | PVT tolerant hybrid current starved ring VCO, bulk-driven keeper, 45nm, AEU | Two current-starved ring VCOs; Cadence GPDK45; corner + Monte Carlo | 6.97GHz, -121.5dBc/Hz@1MHz. PVT tolerance of the VCO itself, no compensation network or loop |
| 15 | S. Madheswaran, R. Panneerselvam, 2024 | 7-stage MCML power-gated current starved VCO for PLL, IJECE | MOS current-mode logic delay cells with power gating, 45nm, simulation | 3.06GHz, -125.9dBc/Hz. Same device class as ours; no closed-loop temperature characterisation |

- Research gap (from all 15): no reported work combines a compensated ring VCO inside a closed loop, characterised across process corners with Monte Carlo rigor -- that gap is what this project fills.

# System Design & Architecture




- Integer-N charge-pump PLL: REF -> PFD -> charge pump -> passive loop filter -> 5-stage current-starved ring VCO -> /8 divider, with a compensation network on the VCO bias path.
- Scoping decisions (deliberate, documented): charge pump is an idealised 20uA behavioural source (transistor-level is Review 3, once layout needs real devices anyway).
- PFD uses edge-detectors + NOR SR-latches, not resettable DFFs -- a reliable async reset fights the latch's own bistable feedback loop.

# Implementation Details

- **Modules developed and integrated**
  - VCO (5-stage current-starved ring), PFD, charge pump, passive loop filter, /8 divider, digital primitive cell library (INV/TG/DFF/TFF) -- all hand-written SPICE, ngspice, headless.
  - Temperature-compensation network: 1st-order (PTAT/CTAT-equivalent) and empirically-refined 2nd-order (curvature-matched), as a parameterised behavioural correction voltage.
  - Acquisition-boosted charge pump (5x current until the divider's first edge) -- closes the cold-start dead-zone risk carried from Review 1.
- **Algorithms implemented**
  - Coefficient derivation: analytical 1st-order (from the measured f(T) polynomial fit and K_VCO), then empirical 2nd-order refit against the measured post-1st-order residual -- zero extra simulation for the refinement step.
  - Statistics: paired t-test, Wilcoxon signed-rank (normal approximation), OLS regression with slope confidence interval -- hand-implemented (no scipy in this environment), self-tested.
- **Code snippet -- the compensation correction (ngspice behavioural voltage source)**

```
Bcomp vctrl vctrl_nom V = 'k1*(T-27) + k2*(T-27)*(T-27)'
  ; k1 = -1.322e-4 V/C, k2 = 2.891e-7 V/C^2 (tt, empirically refined)
  ; verified via an isolated 4-line polarity test before use in the full loop
```

# Results & Analysis (75%)

- VCO tuning: K_VCO ~= 669 MHz/V at 0.9V (polynomial fit, not a secant); 18->358 MHz over 0.65->1.40V. Oscillation lost below ~0.60-0.65V (device-threshold-limited).
- Full 5-corner grid (all of tt/ff/ss/sf/fs, not a 3-corner spot-check) -- corrects the Review-1 finding: real tempco spread is 12.5x, not 3.4x. Worst corner (fs) wasn't checked before.

| Corner | NMOS/PMOS | K_VCO (MHz/V) | Uncompensated tempco |
|---|---|---|---|
| sf | slow / fast | 580.2 | +112 ppm/C (flattest) |
| ff | fast / fast | 763.6 | +332 ppm/C |
| tt | typical / typical | 669.3 | +681 ppm/C |
| ss | slow / slow | 568.2 | +1070 ppm/C |
| fs | fast / slow | 688.7 | +1399 ppm/C (steepest) |

**12.5x**

spread, not the 3.4x previously reported

**669**

MHz/V K_VCO (tt, 0.9V, fit-derivative)

- Sensitivity insight: N/P-mismatched corners (sf, fs) bracket the entire tempco range, not the matched-speed corners (ff, ss) -- points at current-source balance, not absolute device speed, as the driving mechanism.

# Results & Analysis (contd.)

- Compensation refit independently at every corner; every corner improves at both stages. Checked at 4 held-out temperatures never used in fitting (-30/10/40/70C): all stay inside the fitted envelope.

| Corner | Baseline span | 1st-order span | 2nd-order span | 2nd-order max \|dev\| | Meets +/-0.3%? |
|---|---|---|---|---|---|
| ff | 4.15% | 1.06% | 0.10% | 0.07% | Yes |
| sf | 1.40% | 0.42% | 0.24% | 0.15% | Yes |
| tt | 8.51% | 3.05% | 1.12% | 0.87% | No |
| ss | 13.37% | 6.37% | 3.32% | 2.46% | No |
| fs | 17.49% | 9.13% | 5.31% | 3.83% | No |

- **Our +/-0.3% target is met at ff and sf only -- not at the typical corner. Closing the gap needs a transistor-level network whose PTAT slope tracks process, and/or a per-corner trim (Review 3).**
- Naive analytical 2nd-order coefficients made tt worse (4.49% vs 3.05%): they assume constant K_VCO, which is false. Refitting against the measured 1st-order residual fixed it.

*Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.*

# Results & Analysis (contd.) -- Closed-Loop Lock

| Quantity | Coarse-tuned start (0.85V) | TRUE cold start (0V, boosted) |
|---|---|---|
| Lock time | 577 ns | 839 ns |
| Settled V_ctrl | 0.902 V | 0.9002 V |
| Settled f_VCO | 227.96 MHz | 226.42 MHz |
| Spec / prediction | < 2us spec; ~450ns analytical | < 2us spec |

- Cold-start acquisition dead zone (open risk since Review 1) -- fixed and verified. Built a 5x-boosted-current charge pump variant that reverts to nominal current once the divider produces its first edge (tracked via a B-source-charged integrating capacitor, reusing the same current-injection idiom already proven correct -- not a new digital latch and its metastable-initial-state risk).
- Simulated from a TRUE 0V cold start (not the coarse-tuned 0.85V assumption): locks in 839 ns, comfortably inside the 2us spec -- worked on the first simulation attempt.
- Two real bugs found and fixed while closing the original loop (Review 1): an inverted SPICE B-source node-order convention (caught with a 5-line isolated test before touching the full loop), and an inconsistent loop-filter capacitor initial condition (a 50ns charge-pump hold gate was added to test a wrong hypothesis first -- disproving it is what pointed at the real cause).

*Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.*

# Results (contd.) -- Monte Carlo & Benchmark

| Arm (tt, mismatch on, paired by seed) | n | Mean \|drift\|, 85C vs -40C | 95% CI |
|---|---|---|---|
| Baseline (uncompensated) | 12 | 18.93 MHz (8.4%) | [16.20, 21.65] |
| 1st-order | 30 | 5.86 MHz (2.6%) | [4.62, 7.09] |
| 2nd-order (curvature-matched) | 30 | 3.37 MHz (1.5%) | [2.33, 4.41] |

- **Both stages significantly reduce drift (12/12 seeds). 2nd-order beats 1st-order: -2.48 MHz, 95% CI [0.81, 4.15], paired t-test and Wilcoxon agree, 23/30 seeds.**
- Correction on record: our first pass tested signed drift and overstated this at n=12 (1st-order over-corrects, so the sign flipped). On |drift| it was not significant at n=12 -- n=30 is what earns the claim.

| Design | Process | Residual drift | Lock time |
|---|---|---|---|
| This work, 2nd-order (tt) | sky130, simulation | +/-0.87% max dev., -40 to 85C | 839 ns (0V cold start) |
| Souliotis et al. 2022 [1] | 180nm, measured silicon | -1.5% / +0.4%, -40 to 120C | not reported |
| Stancu et al. 2023 [2] | simulation, 50 MHz clock | 1.6% temperature error | n/a |
| Geng et al. 2018 [3] | 180nm, measured silicon | not reported as drift | < 2 us |

- Comparable to published work, but ours is simulation of an ideal correction network vs. their measured silicon -- not a like-for-like claim until the transistor-level network exists.

*Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.*

# Challenges & Remaining Work

- **Challenges faced and solutions**
  - SPICE B-source polarity and an inconsistent loop-filter initial condition broke the first closed-loop runs -- each isolated with a minimal test (and one wrong hypothesis disproved) before touching the full loop.
  - Naive analytical 2nd-order coefficients over-corrected the cold end -- fixed by refitting against the measured 1st-order residual; the refit was then checked at 4 held-out temperatures it never saw.
  - Caught our own statistics error: the first Monte Carlo pass compared SIGNED drift, which made 1st- vs 2nd-order look significant only because the sign flipped. Re-tested on |drift| and re-run at n=30.
  - Novelty check found a 2017 patent on generic quadratic compensation -- narrowed our claim to ring-VCO + closed-loop PLL + open PDK + statistically validated quantification.
- **Remaining work (25%)**
  - Close the spec gap: +/-0.3% is met at ff and sf only (tt 0.87%, ss 2.46%, fs 3.83%) -- a transistor-level compensation network whose PTAT slope tracks process, plus a per-corner trim.
  - Transistor-level charge pump; layout in Magic, DRC, LVS (netgen), post-layout re-simulation.
  - Cadence/Virtuoso cross-validation arm and the actual H1 regression (eps_res vs C) -- the other half of the research hypothesis, pending lab access.
- **Timeline to Review-III (28.10.2026)**
  - Weeks 1-2: transistor-level CP + compensation network.  Week 3: layout / DRC / LVS.  Week 4: Virtuoso arm, H1 regression, final report.

# References


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- [3] Z. Geng, X. Ma, and Z. Li, “A low power fast locking PLL frequency synthesizer with temperature compensation,” in Proc. 3rd Int. Conf. EAME, Atlantis Press, Jun. 2018, pp. 281–285.
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# Thank You

Project-I (2026) · SENSE · VIT Chennai · github.com/pocopepe/visualizer